

Question No:6

Problem Description

Jenish and Neha are excellent friends. After performing several queries, Neha challenges Jenish to determine the highest possible Rating of the provided array A . According to Neha, the highest occurrence of an element in an array is its rating.

Jenish is given M and Q , the Magical Numbers. Jenish may perform Addition or Subtraction with M at most Q times for each element in the supplied Array A .

Because Jenish is stumped and unable to discover a solution, assist him in determining the highest possible ratings for the given array after applying queries to each element.

Constraints :

- $1 \leq N \leq 1000000$
- $1 \leq M \leq 100$
- $1 \leq Q \leq 10$
- $1 \leq A_i \leq 1000000$

Input :

- First line of Input contains integer M
- Second line contains integer Q
- Third line contains integer N
- Fourth line contains N elements representing elements of array A

Output :

- Output the highest possible rating of array A after applying Queries.

Explanation for test case 1

Jenish can add 1 in 1st element and subtract 1 from 3rd element to get highest frequency of 2 , i.e. 3

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
1
1
4
1 2 3 4
```

EXPECTED OUTPUT

```
3
```

Test Case 2

INPUT (STDIN)

```
1
1
5
11 12 31 41 51
```

EXPECTED OUTPUT

```
2
```

Code:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int M, Q;
5
6  int compare(const void *a, const void *b) {
7      int x = *(int *)a;
8      int y = *(int *)b;
9
10     int r1 = x % M;
11     int r2 = y % M;
12
13     if (r1 != r2)
14         return r1 - r2;
15
16     return x - y;
17 }
18
19 int main() {
20     int N;
21     scanf("%d", &M);
22     scanf("%d", &Q);
23     scanf("%d", &N);
24
25     int A[N];
26
27     for (int i = 0; i < N; i++) {
28         scanf("%d", &A[i]);
29     }
```

```

30
31     qsort(A, N, sizeof(int), compare);
32
33     int maxRating = 0;
34     int left = 0;
35
36     for (int right = 0; right < N; right++) {
37
38         while (left <= right &&
39             (A[left] % M != A[right] % M ||
40             (A[right] - A[left]) / M > 2 * Q)) {
41             left++;
42         }
43
44         int count = right - left + 1;
45
46         if (count > maxRating) {
47             maxRating = count;
48         }
49     }
50
51     printf("%d\n", maxRating);
52
53     return 0;
54 }

```

Output:

Test Case-1:

	<pre> CSE401\DSA\DSA_PRACTICE\Prac 1 1 4 1 2 3 4 3 PS C:\Users\kesha\OneDrive\ </pre>	
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Test Case-2:

	<pre> CSE401\DSA\DSA_PRACTICE\Prac 1 1 5 11 12 31 41 51 2 PS C:\Users\kesha\OneDrive\ </pre>	
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